

3. Hospital Patient Admissions

Our regional hospital system needs to analyze patient flow and resource utilization to improve care quality and operational efficiency. When a patient is admitted, we track them from admission to discharge. Each patient is assigned to a specific department (e.g., 'Cardiology', 'Orthopedics', 'General Surgery') and is under the care of a primary attending physician. We also record one or more diagnoses for each admission. We need to answer critical questions like: What is the average length of stay for a patient, and how does it differ by department or primary diagnosis? We also track the total cost of treatment for each admission. We are particularly interested in readmission rates—what percentage of patients are readmitted within 30 days of discharge? We want to see how these metrics trend over time (monthly and quarterly). We also need to analyze physician workload by looking at the number of patients they admit, and compare treatment costs across different doctors for the same diagnosis.

Current OLTP

